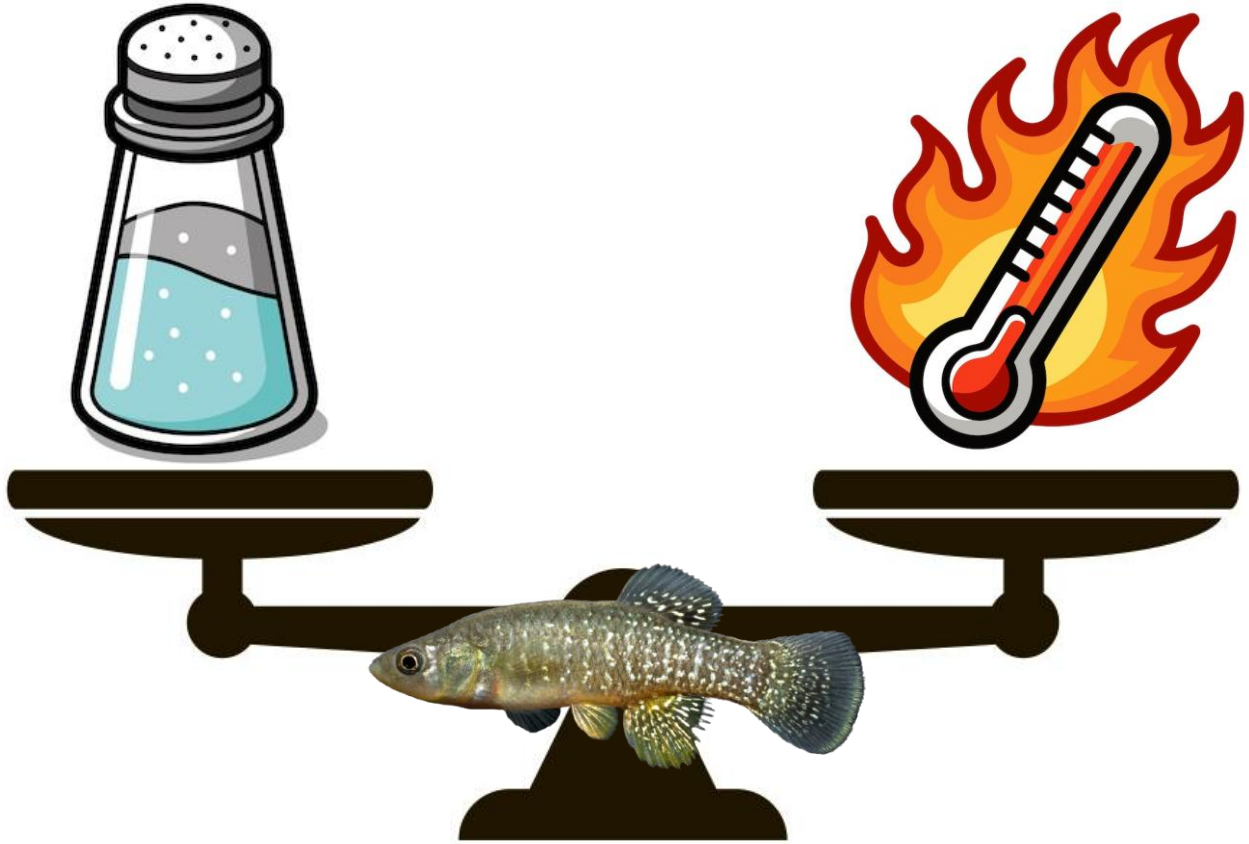
Metabolic costs of Multi-stressor environments

Overcoming the Environment Requires Energy



Overlap in Metabolic Responses

Temperature

↑ metabolism in cold

↓ metabolism in heat

Salinity

↑ metabolism for cost of osmoregulation

- Is there also an overlap in signalling

Predictions

High salinity × High temperature

Isosmotic salinity × Low temperature

=

Undermining response

Hypothesis: The cost of osmoregulation will compromise thermal acclimation

3-Factor Experimental Design

3-week acclimations

1. Temperature Acclimation

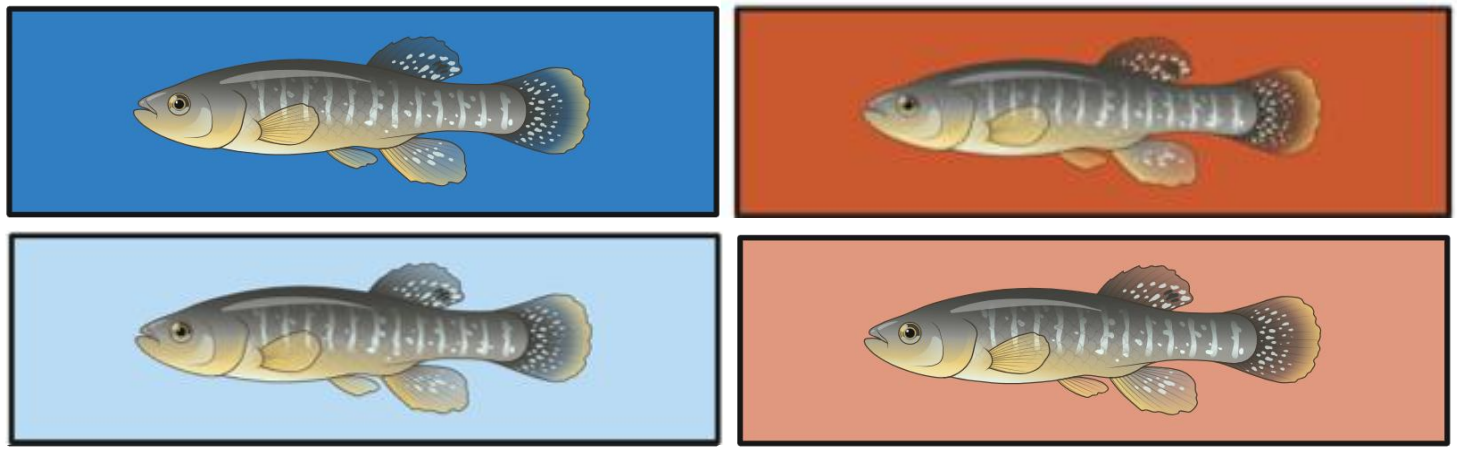
10°C

24°C

28ppt

2. Salinity Acclimation

11ppt
(isosmotic)



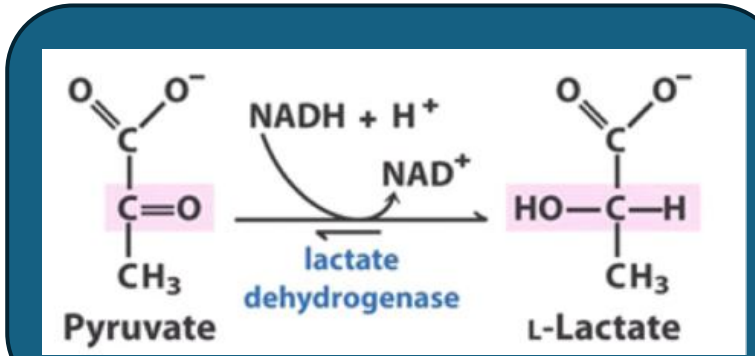
3. Assay Temperature

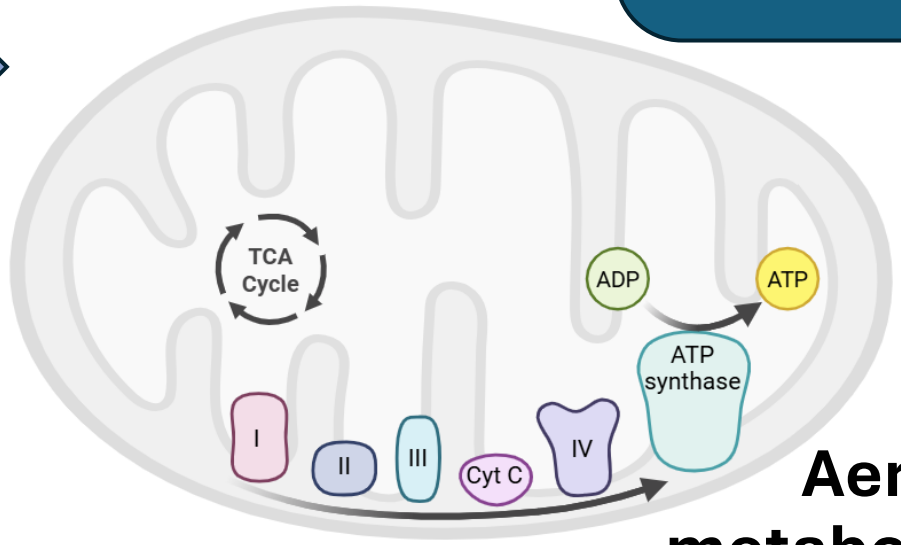
10°C

24°C

Measure Mitochondrial Enzymes

Anaerobic metabolism





Aerobic metabolism

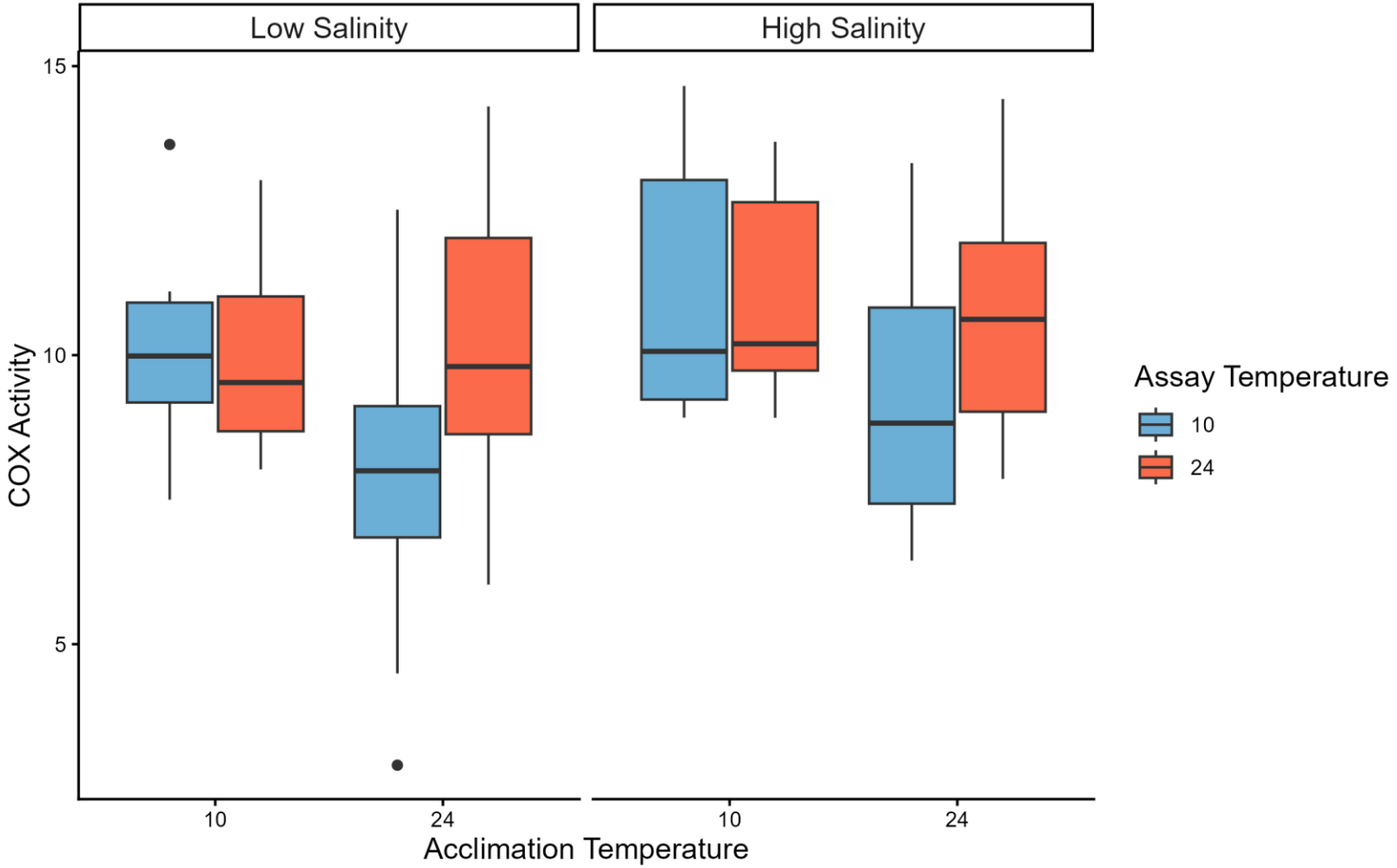
What does the data look like?

- Predictor Variables:
  - Acclimation Temperature
  - Acclimation Salinity
  - Assay Temperature
- Response Variables
  - COX Activity
  - CS Activity
  - LDH Activity
  - *Body Condition*

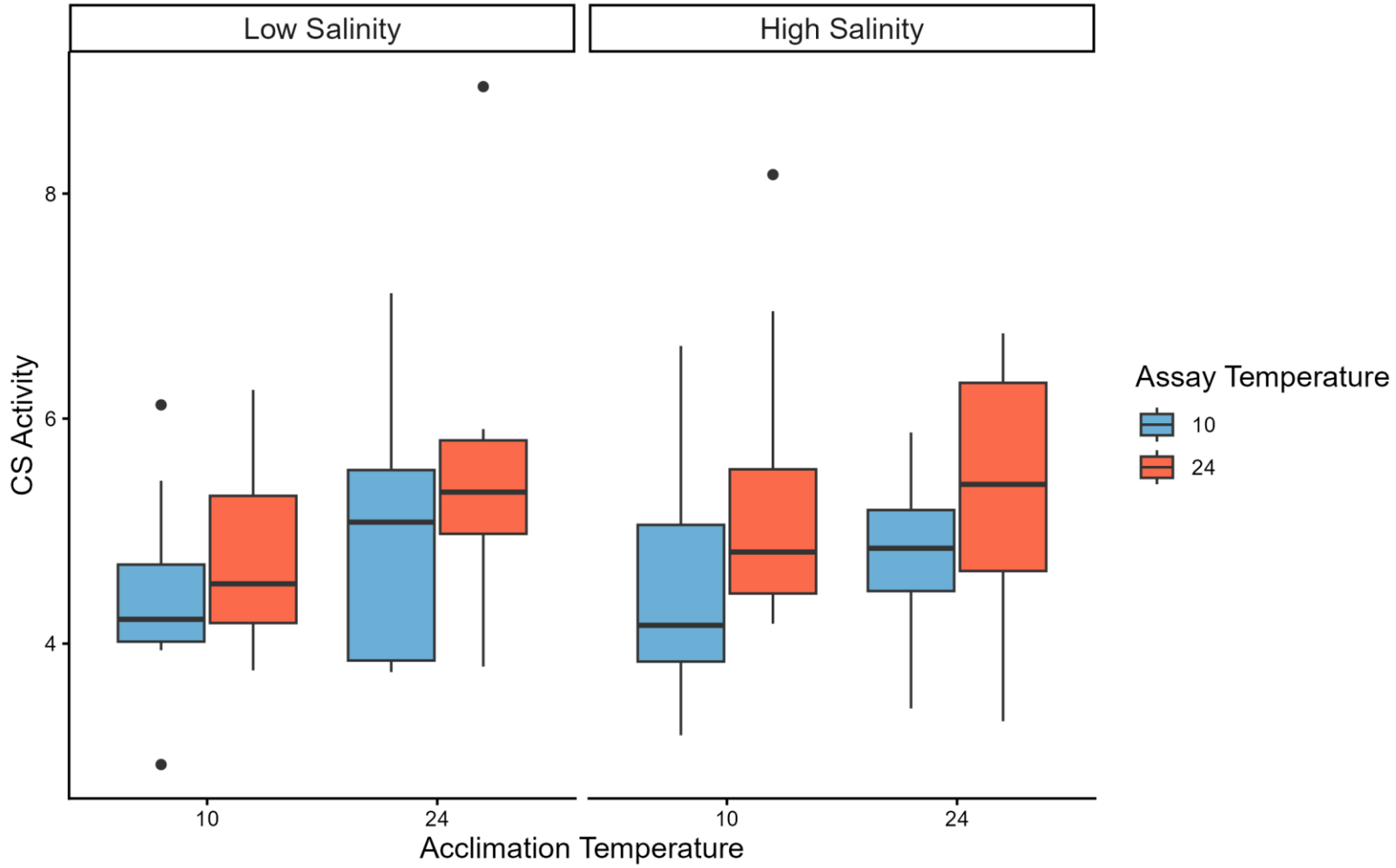
	Fish_ID	Total_Length	Mass	Accliamtion_Temperature	Acclimation_Salinity	Assay_Temperature	COX_Activity	CS_Activity	LDH_Activity	Hematocrit	Body_Condition
1	74	74	4.735	10	11	10	13.645	5.448	128.559	33.33	1.168490
2	74	74	4.735	10	11	24	13.028	6.020	172.951	33.33	1.168490
3	75	65	3.586	10	28	10	14.657	6.145	134.337	NA	1.305781
4	75	65	3.586	10	28	24	13.124	6.955	186.454	NA	1.305781
5	76	66	3.203	24	11	10	12.517	7.115	156.454	NA	1.114102
6	76	66	3.203	24	11	24	14.303	8.951	207.955	NA	1.114102
7	77	69	3.894	24	28	10	9.836	5.035	114.277	35.71	1.185356
8	77	69	3.894	24	28	24	10.716	5.394	143.404	35.71	1.185356
9	78	70	4.403	10	11	10	11.102	4.177	110.235	25.93	1.283673
10	78	70	4.403	10	11	24	11.774	5.078	149.688	25.93	1.283673
11	79	79	6.167	10	28	10	8.919	3.185	78.879	29.41	1.250814
12	79	79	6.167	10	28	24	8.917	5.066	112.277	29.41	1.250814
13	80	74	4.994	10	28	10	14.627	6.646	134.980	37.50	1.232405
14	80	74	4.994	10	28	24	13.691	8.169	182.689	37.50	1.232405
15	81	61	2.773	24	11	10	7.849	5.505	117.334	30.00	1.221688
16	81	61	2.773	24	11	24	9.314	5.907	151.866	30.00	1.221688
17	82	69	3.871	24	28	10	12.224	5.644	129.346	NA	1.178354
18	82	69	3.871	24	28	24	12.301	6.458	156.354	NA	1.178354

Showing 1 to 19 of 64 entries, 11 total columns

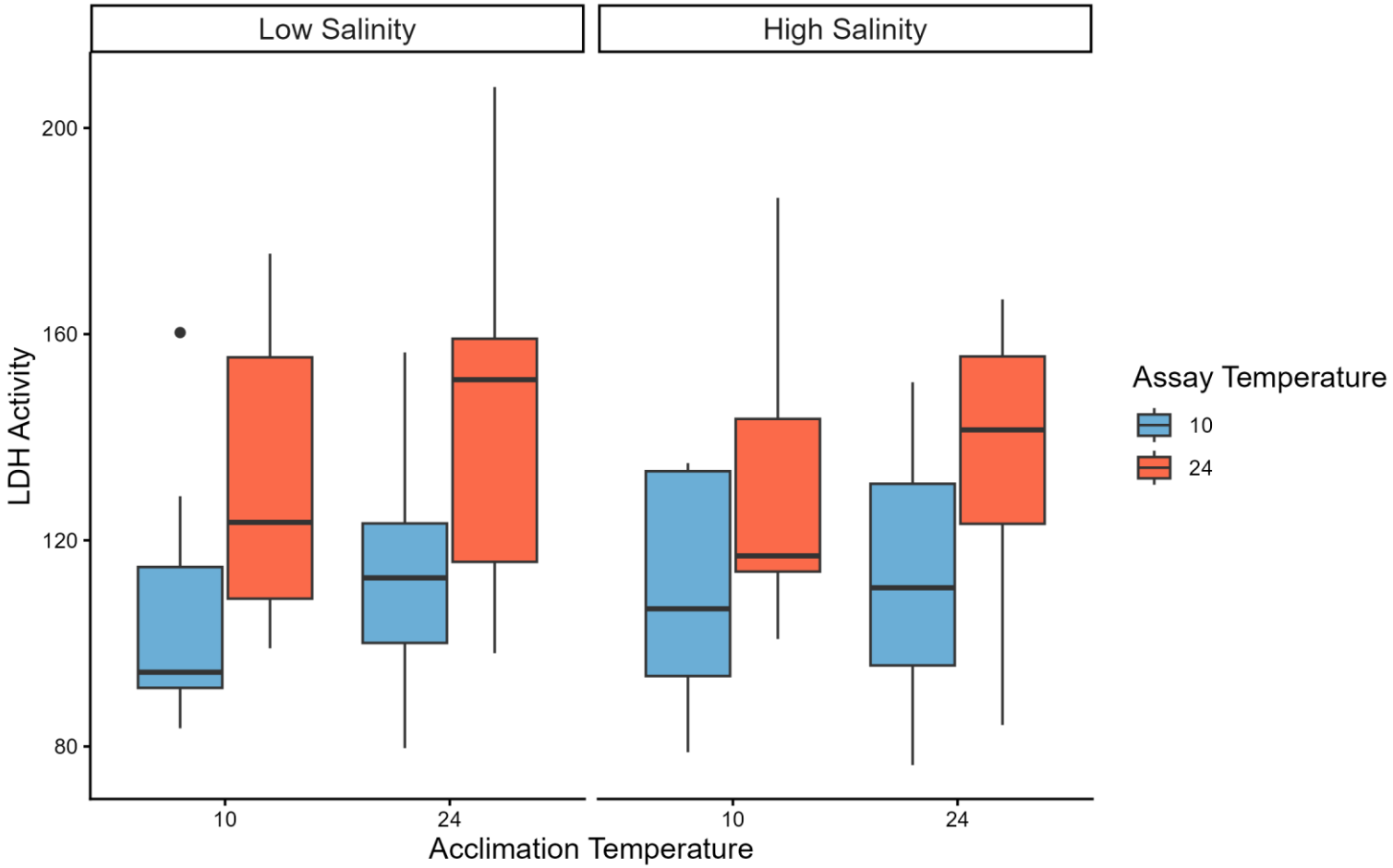
Data Visualization: COX Activity



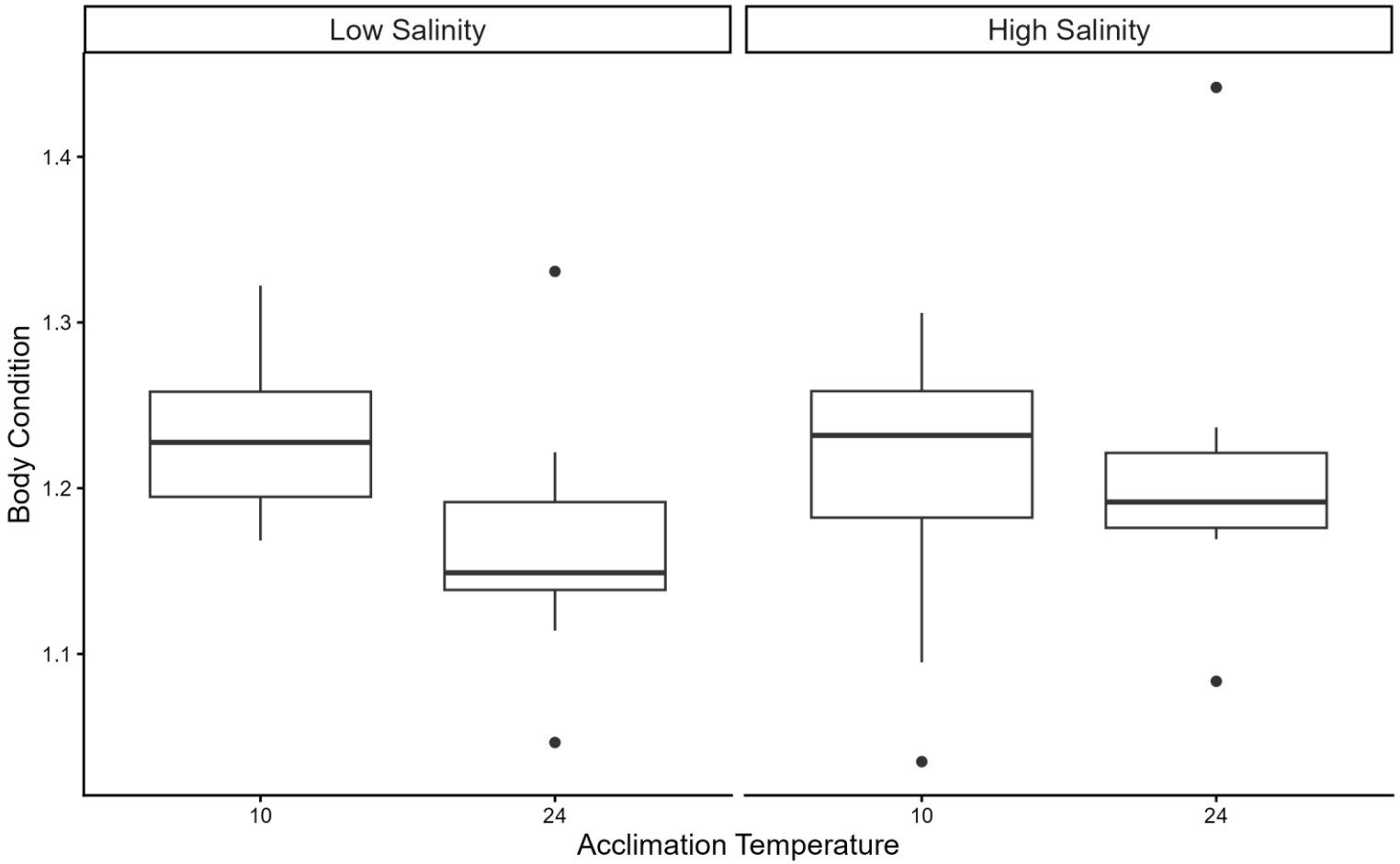
Data Visualization: CS Activity



Data Visualization: LDH Activity



Data Visualization: Body Condition



Statistical Model Choice

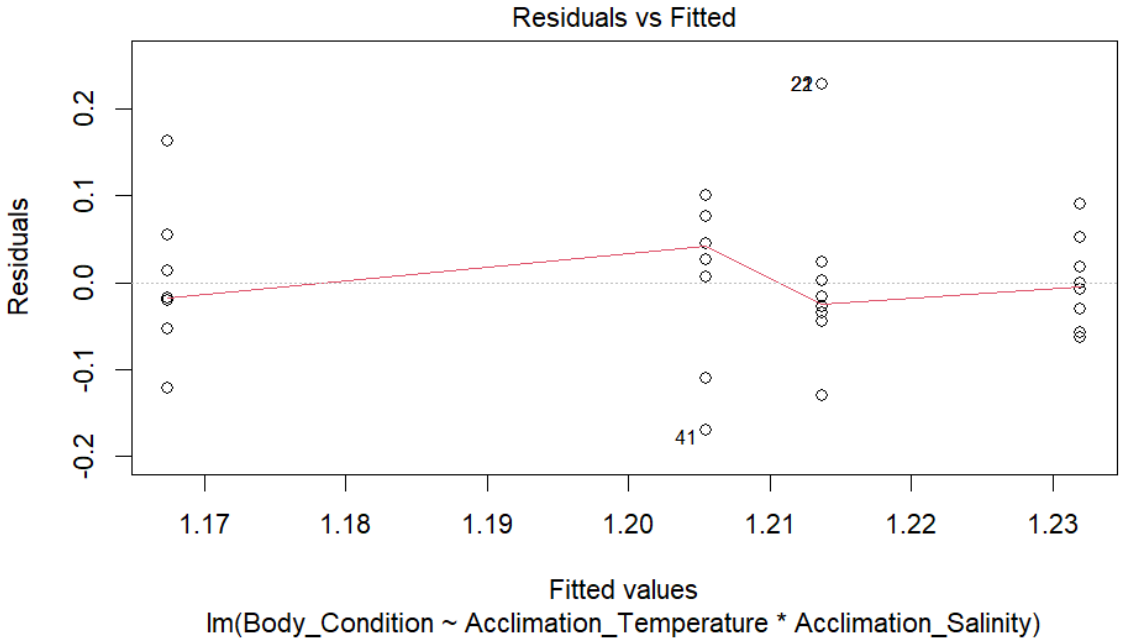
Linear model for body condition

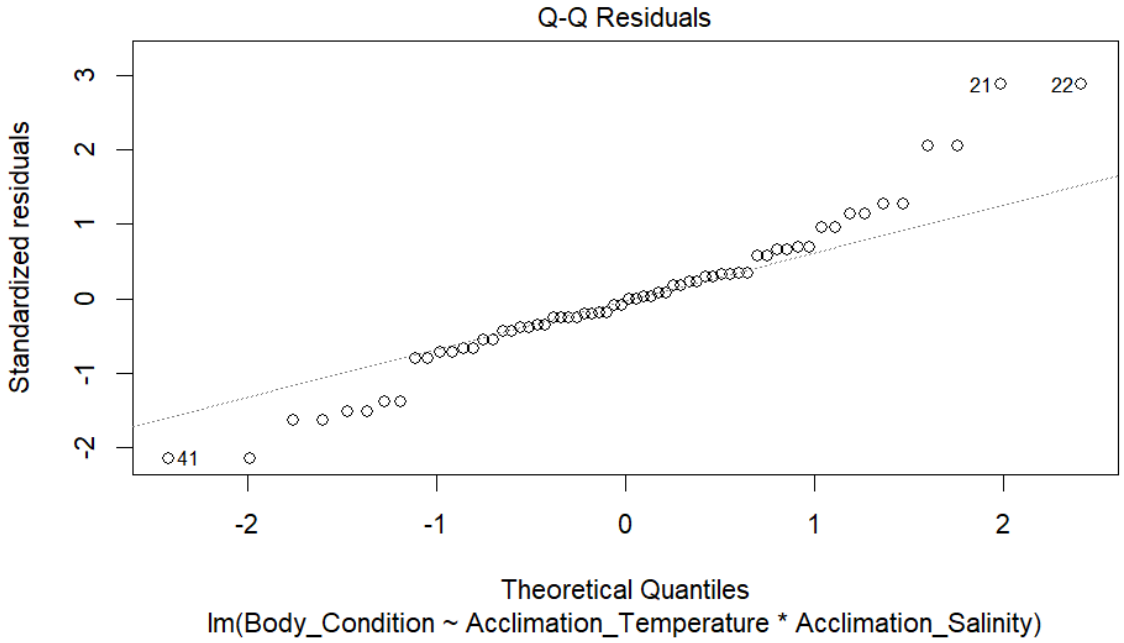
Manova for enzyme activity

- Response variables are not independent
- Categorical predictor variables
- Continuous response variables

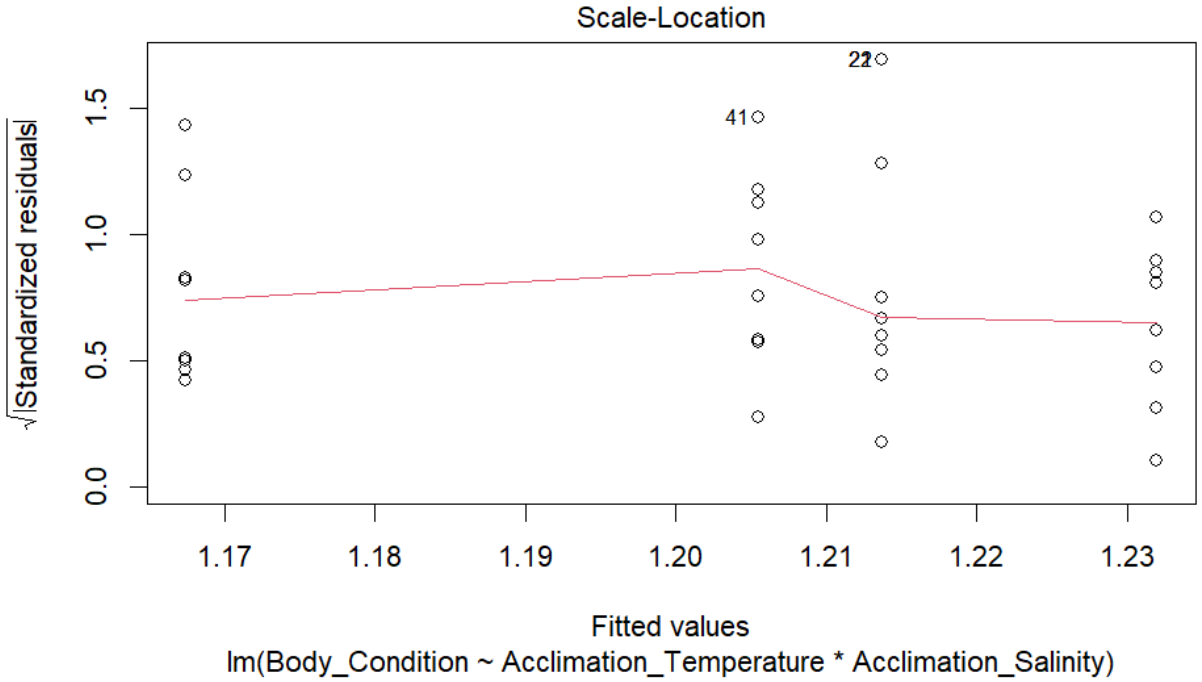
Linear Model: Body Condition

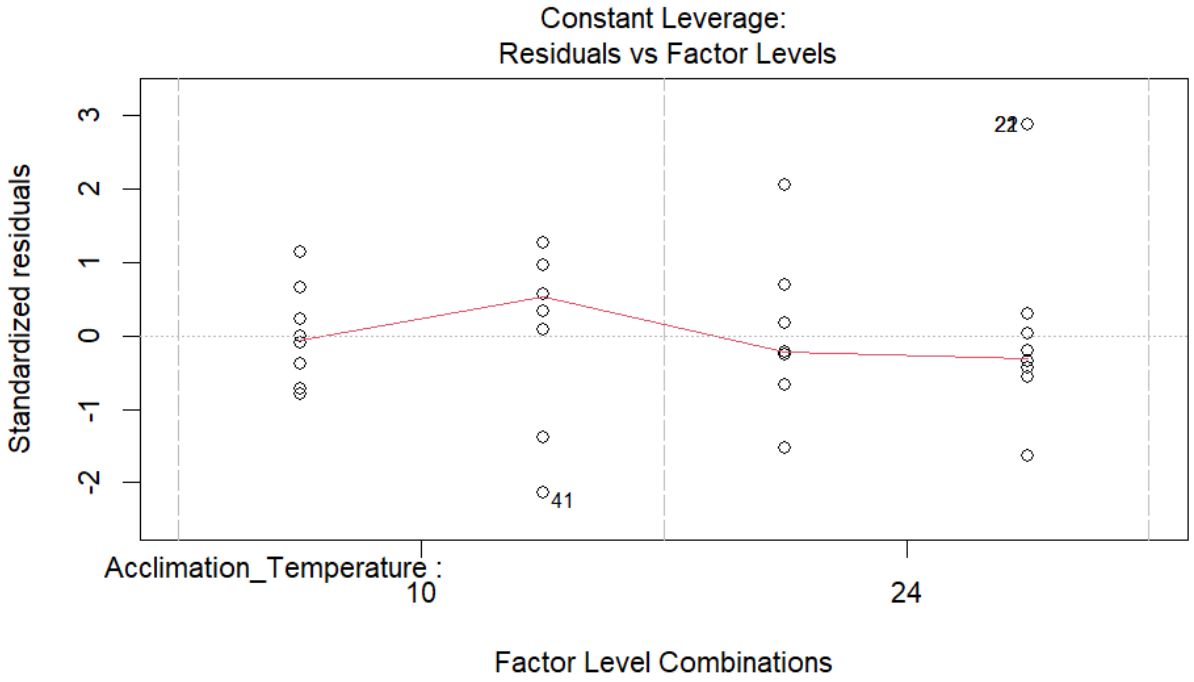
Interaction between acclimation temperature and salinity





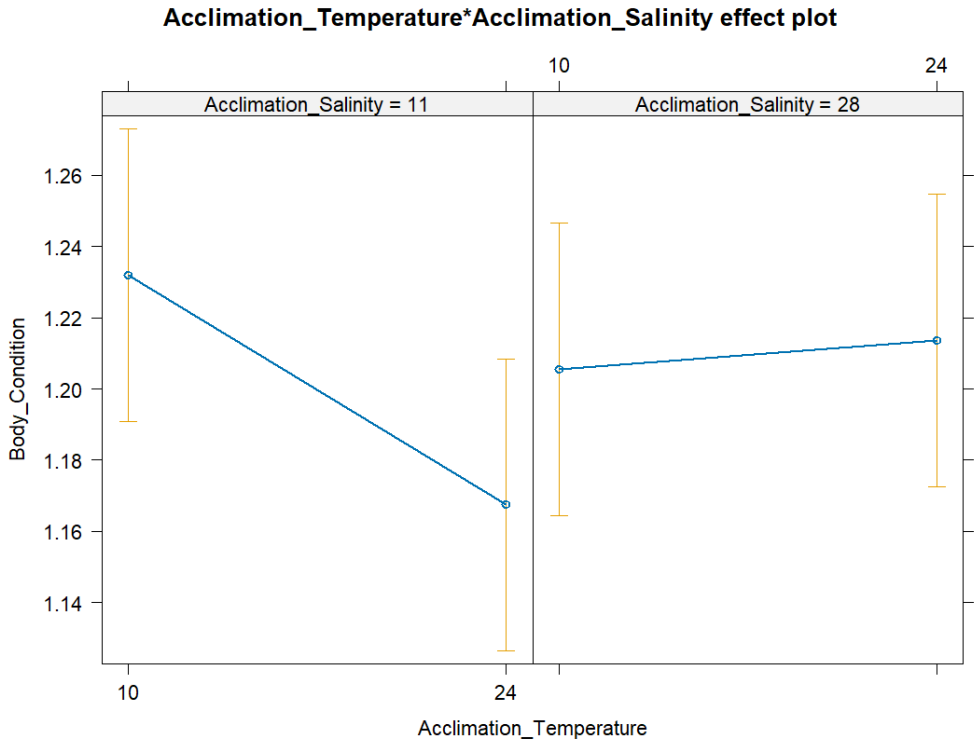
Diagnostic plots





Linear Model

```
Coefficients:
                                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                                      1.23196    0.02053  60.005   <2e-16 ***
Acclimation_Temperature24                       -0.06453    0.02904  -2.222   0.0300 *
Acclimation_Salinity28                          -0.02653    0.02904  -0.914   0.3645
Acclimation_Temperature24:Acclimation_Salinity28  0.07274    0.04106   1.771   0.0816 .
---
```

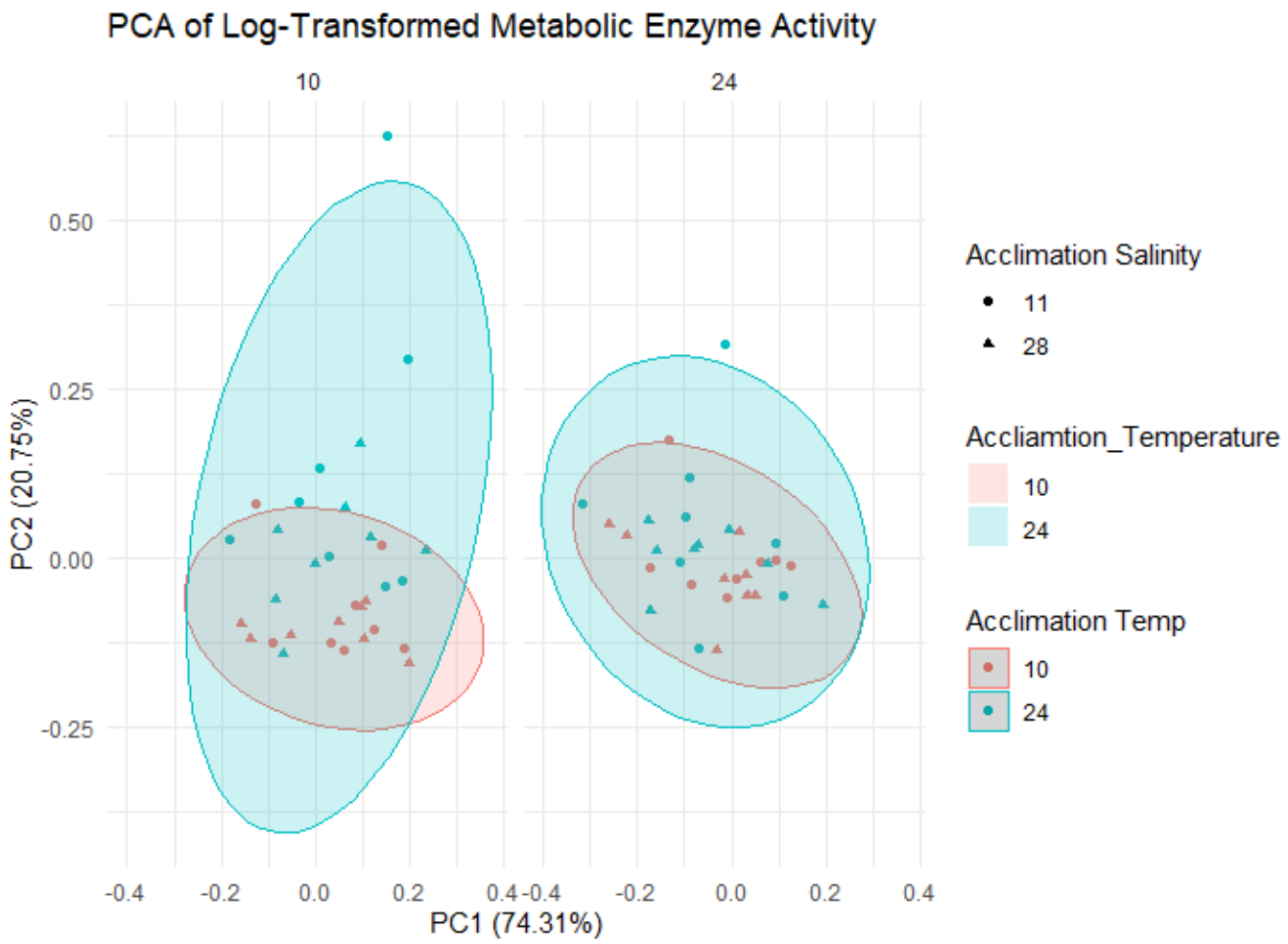


MANOVA

- Interaction of acclimation temperature and salinity and assay temperature influenced the three enzyme activites

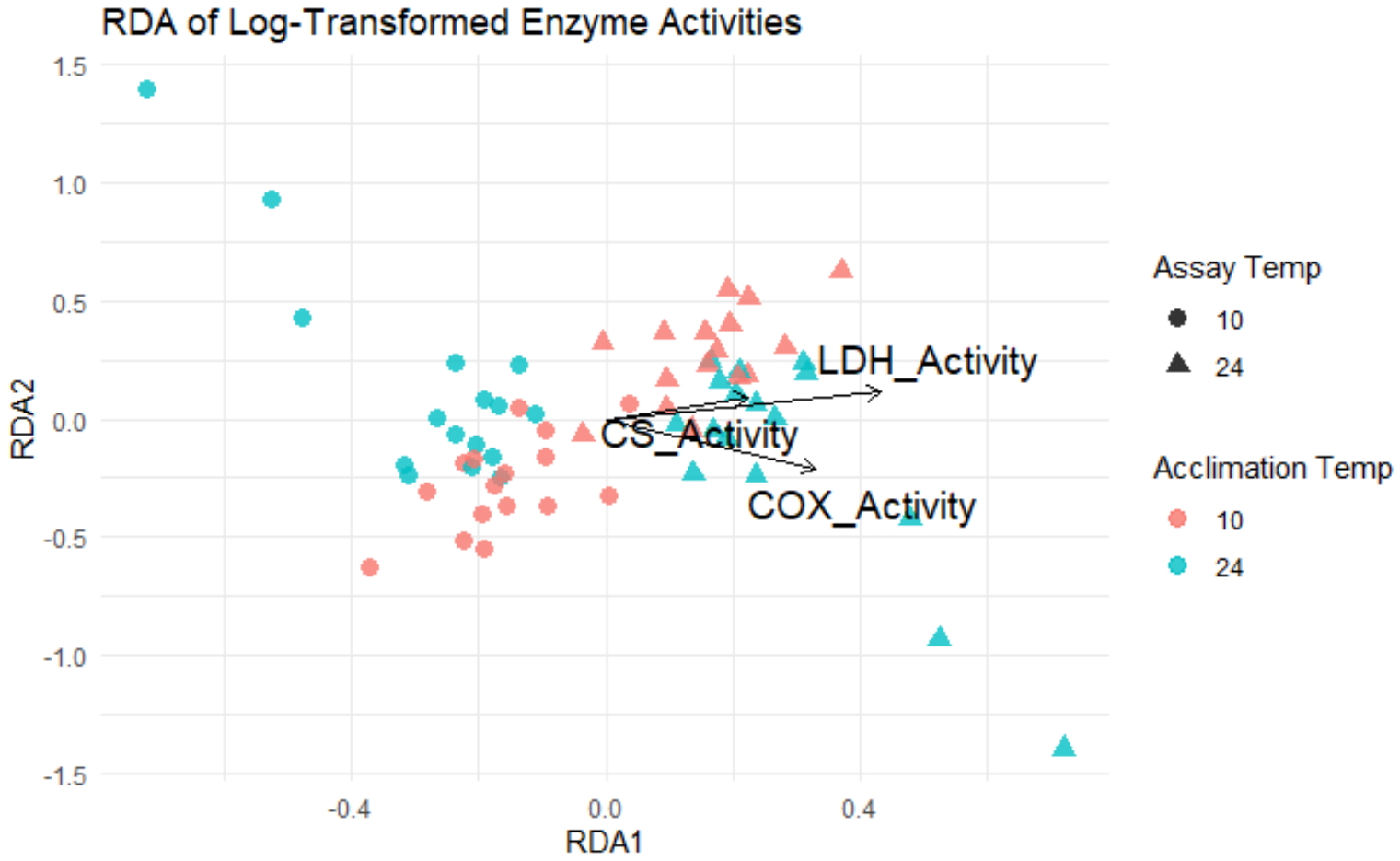
```
Type II MANOVA Tests: Pillai test statistic
                                                                Df test stat approx F num Df den Df    Pr(>F)
Acclimation_Temperature                                          1  0.207292   4.7070      3     54 0.0054396
Acclimation_Salinity                                             1  0.096672   1.9263      3     54 0.1362571
Assay_Temperature                                                1  0.261053   6.3590      3     54 0.0009027
Acclimation_Temperature:Acclimation_Salinity                     1  0.038231   0.7155      3     54 0.5470198
Acclimation_Temperature:Assay_Temperature                        1  0.095986   1.9112      3     54 0.1387109
Acclimation_Salinity:Assay_Temperature                           1  0.031879   0.5927      3     54 0.6224600
Acclimation_Temperature:Acclimation_Salinity:Assay_Temperature   1  0.008948   0.1625      3     54 0.9211219
```

Data Exploration




```
Importance of components:
                          PC1    PC2     PC3
Standard deviation     1.4931 0.7890 0.38479
Proportion of Variance 0.7431 0.2075 0.04935
Cumulative Proportion  0.7431 0.9506 1.00000
                    PC1        PC2         PC3
COX_Activity -0.4851496 -0.8735373 -0.03952819
CS_Activity  -0.6145284  0.3727615 -0.69527239
LDH_Activity -0.6220809  0.3130199  0.71765857
```

Data Exploration




```
Accumulated constrained eigenvalues
Importance of components:
                          RDA1     RDA2      RDA3
Eigenvalue             0.01876 0.003566 0.0002089
Proportion Explained   0.83245 0.158275 0.0092706
Cumulative Proportion  0.83245 0.990729 1.0000000
```

Next steps

- Produce diagnostics for the RDA and MANOVA
- Look into the effect size
  - We see there is correlation in variation but is the biologically meaningful
- Adding in Fish_ID as a random effect
- Look into the response variable hematocrit